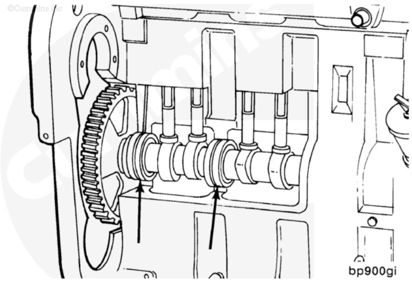
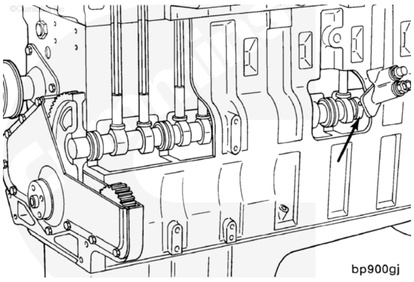
Trainings

General Information

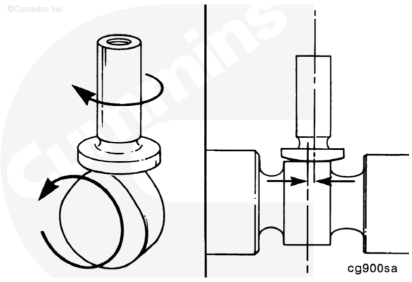
The camshaft is gear-driven from the crankshaft. A replaceable bushing is used for each of the camshaft journals.



The camshaft has lobes to operate the intake and exhaust valves. The valve lobes contact the valve tappets, which operate the valves. The profile of the camshaft lobes is the same for all C Series engines.

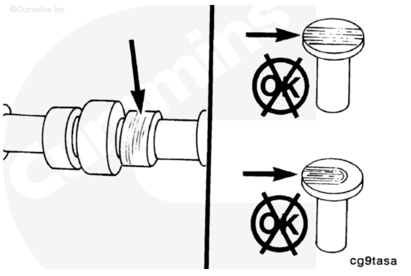


The tappets are mushroom-shaped. The offset position of the tappet against the camshaft lobe causes the tappet to rotate as it lifts the push rod.



Diagnosing Malfunctions

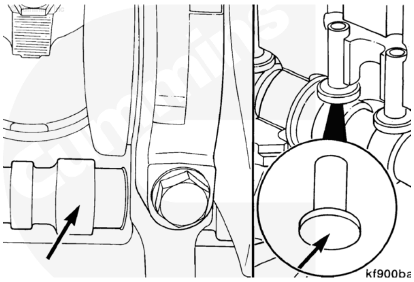
Loose rocker levers and the need to reset the valve clearance frequently can indicate camshaft lobe or tappet wear. If an inspection of the levers, valve stems, and push rods does **not** reveal wear, tappet or camshaft lobe wear should be suspected.



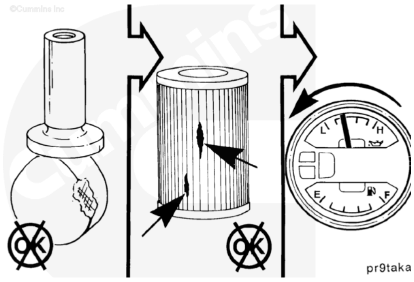
⚠ CAUTION ⚠

To reduce the possibility of engine damage, any time a new camshaft is installed new tappets and push rods must also be installed.

The camshaft lobes can be inspected after removing the oil pan. Similarly, the faces of the tappets can be inspected after removing the push rods and lifting the tappets.



A severely damaged camshaft journal(s) can generate small metal particles that can be found in the oil pan and oil filter. These metal particles will also be indicated as iron in an oil analysis. As the clearance in the journal(s) increases, a small decrease in oil pressure can be detected.



The camshaft end clearance is determined by the clearance between the camshaft and the thrust plate. The camshaft gear **must** be removed to adjust the camshaft end clearance.

Note : The camshaft does not have to be removed to remove the camshaft gear. Use the camshaft gear puller, Part Number 3376400.

Camshafts that are damaged or worn on the injector or valve lobes **must** be replaced. Cummins Inc. does **not** recommend grinding camshaft lobes.

Preparatory Steps

- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/100/100-003-011.html)
- Remove the rocker levers. Refer to Procedure 003-008 in Section 3. (/qs3/pubsys2/xml/en/procedures/100/100-003-008.html)
- Remove the push rods. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/100/100-004-014.html)
- Remove the drive belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/100/100-

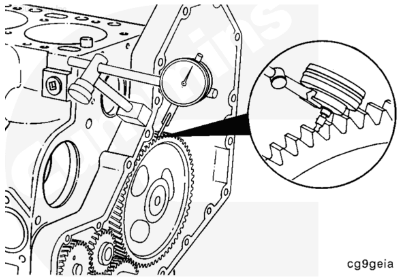
008-002.html)

- Remove the vibration damper. Refer to Procedure 001-052 in Section 1. (/qs3/pubsys2/xml/en/procedures/100/100-001-052.html)
- Remove the front gear cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/100/100-001-031.html)

Initial Check

Note : If the camshaft or camshaft gear is replaced due to gear train noise, check the gear train backlash. Excessive gear backlash can be caused by worn camshaft bushings.

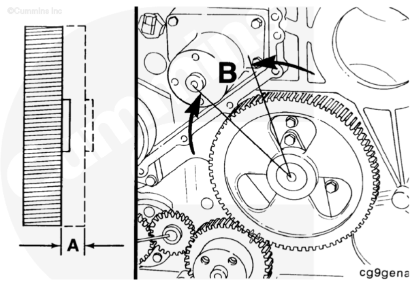
Position a dial indicator on a tooth of the camshaft gear.



Measure the camshaft end clearance and backlash.

Camshaft End Clearance (A)		
mm		in
0.12	MIN	0.005
0.46	MAX	0.018

Camshaft Gear Backlash (B)		
mm		in
0.08	MIN	0.003
0.33	MAX	0.013



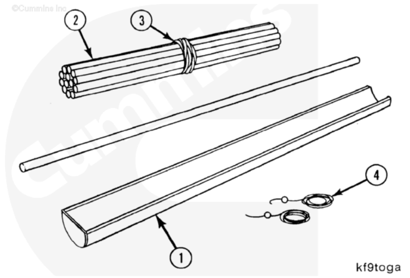
Note : Mark the camshaft gear and crankshaft gear for further analysis if the backlash or end clearance exceeds limits.

Remove

The tappet removal tool kit, Part Number 3822513, contains:

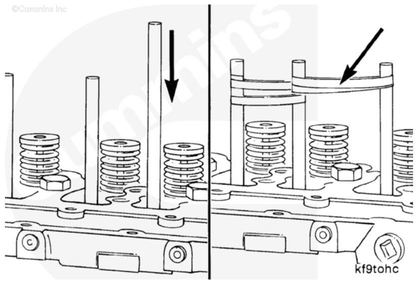
1. Valve tappet tray
2. Dowel rods
3. Rubber bands
4. Nylon string.

Note : This kit is used to lift the tappets so the camshaft can be removed.

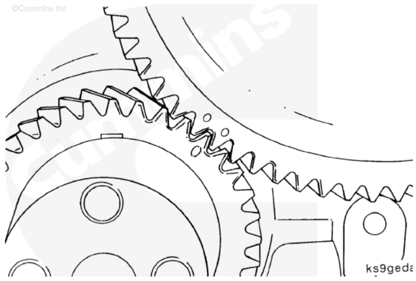


Press a wooden dowel rod into each tappet. It will probably be necessary to push the dowel into the tappet with a soft-face hammer.

Pull each valve tappet up until it makes contact with the cylinder block. Place a rubber band around two dowels. This will hold the tappets up off the camshaft.

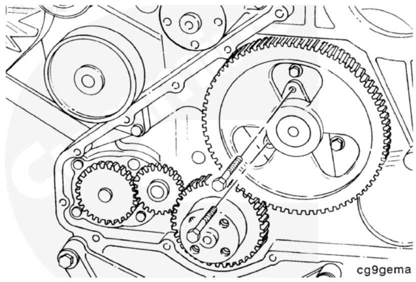


Rotate the crankshaft to align the timing marks before removing the camshaft.



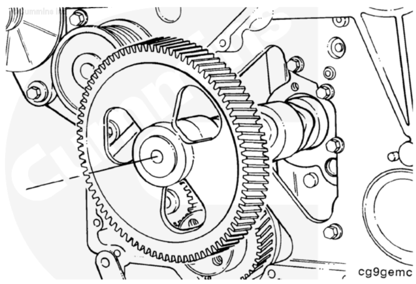
Remove the camshaft thrust plate capscrews.

Remove the camshaft thrust plate.



Remove the camshaft from the cylinder block.

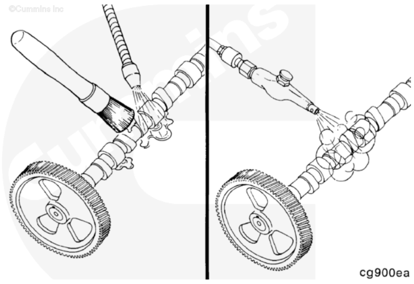
Note : Rotate the camshaft as it is being removed. Use extreme care to make sure that the camshaft bushings are not damaged during the camshaft removal process.



Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

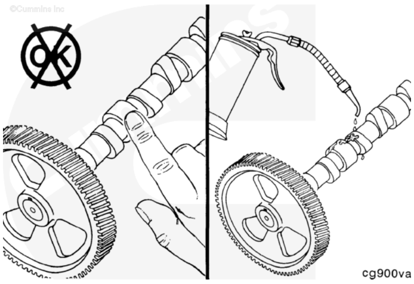
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

Clean the camshaft with steam or solvent.

Dry the camshaft with compressed air.

⚠ CAUTION ⚠

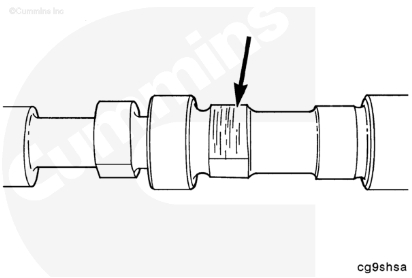
After the camshaft has been cleaned, do not touch the machined surfaces with bare hands. This will cause rust to form.



Lubricate the camshaft with clean engine oil before handling it.

Inspect for Reuse

Note : If a new camshaft is installed on an engine that uses slider tappets, new tappets and push tubes **must** also be installed. If a new camshaft is installed on an engine that uses roller tappets, **only** the damaged roller tappets **must** be replaced.

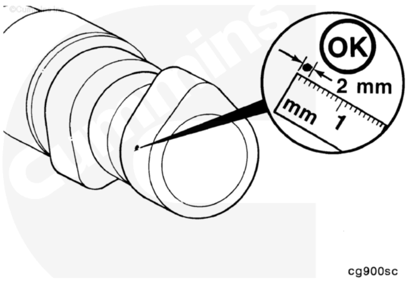


Inspect the valve lobes and bearing journals for wear, cracking, pitting, scratches, and other damage.

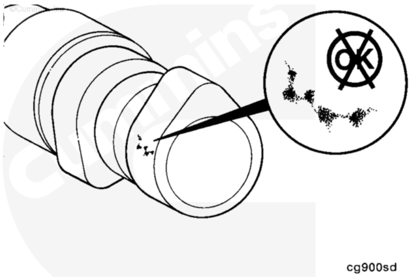
The following criteria define the size of the pits, allowable wear, and edge deterioration of the camshaft lobes.

Pitting Reuse Criteria

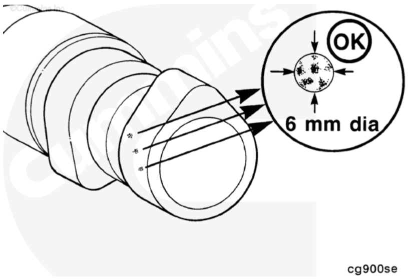
A single pit **must not** be greater than the area of a 2.0 mm [0.079 in] diameter circle.



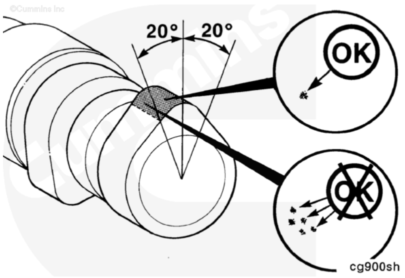
Interconnection of pits is **not** allowable and is treated as one pit.



The total pits, when added together, **must not** exceed a circle of 6.0 mm [0.236 in].

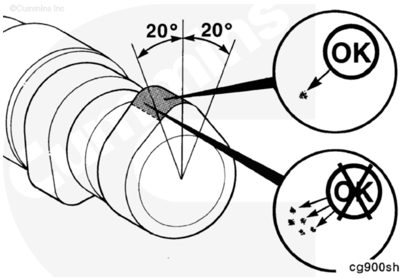


Only one pit is allowed within ±20 degrees of the nose of the cam lobe.

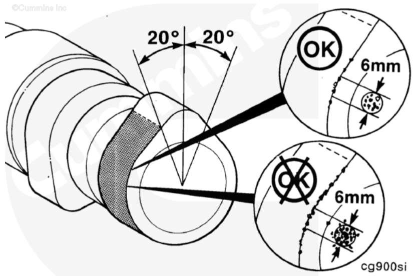


Edge Deterioration (Breakdown) Criteria

The area of edge deterioration can **not** be greater than the equivalent area of a 2.0 mm [0.079 in] circle within ±20 degrees of the nose of the cam lobe.



Outside ±20 degrees of the nose of the cam lobe, the areas of edge deterioration can **not** be greater than the equivalent area of a 6 mm [0.236 in] circle.

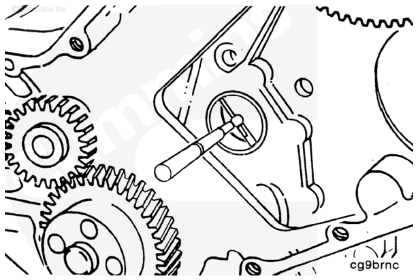


Measure the camshaft bushing bore. Use a bore gauge. The bores **must** be measured in two positions. Take a second measurement 90 degrees from the first measurement.

Camshaft Bushing Bore Inside Diameter				
	mm		in	
Bushing		60.12	MAX	2.367

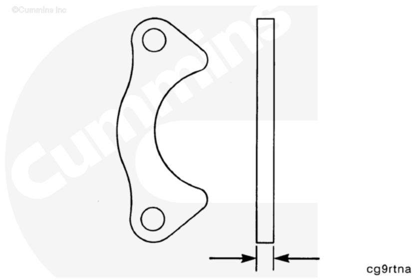
If the bushing diameter is more than 60.12 mm [2.367 in], the camshaft bushings **must** be replaced.

Note : The engine must be removed for camshaft bushing replacement. Refer to Procedure 000-001 in Section 0. (/qs3/pubsys2/xml/en/procedures/41/41-000-001.html)



Measure the camshaft thrust plate thickness.

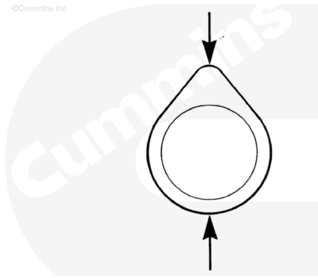
Camshaft Thrust Plate Thickness		
mm		in
9.34	MIN	0.368
9.58	MAX	0.377



Measure

Valve Lobe Wear Criteria

Measure across the peak of the valve lobe.



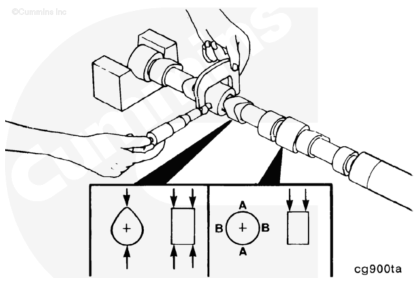
cg900sg

6C - Camshaft Lobe Wear Limits for Part Numbers 3924471, 3923388, 3923478, and 3934168			
	mm		in
Intake	49.940	MIN	1.966
Exhaust	48.916	MIN	1.926

6C - Camshaft Lobe Wear Limits for Part Numbers 3921176, 3914640, and 3919799			
	mm		in
Intake	49.940	MIN	1.966
Exhaust	49.774	MIN	1.960

6C - Camshaft Lobe Wear Limits for Part Number 3930347			
	mm		in
Intake	45.400	MIN	1.787
Exhaust	44649	MIN	1.758

6C - Camshaft Lobe Wear Limits for Part Number 3927693			
	mm		in
Intake	45.400	MIN	1.787
Exhaust	45.141	MIN	1.777



cg900ta

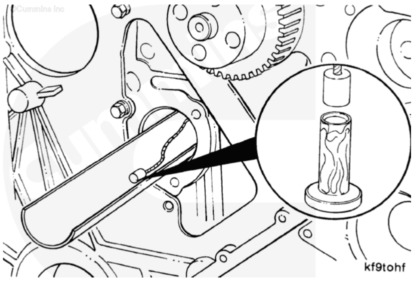
Camshaft Journal Diameter Wear Limits for Part Numbers 3924471, 3923388, 3921176, 3923478, 3914640, 3919799, and 3934168		
mm		in
59.967	MIN	2.3609
60.013	MAX	2.3627

Camshaft Journal Diameter Wear Limits for Part Numbers 3930347 and 3927693		
mm		in
53.967	MIN	2.1247
54.013	MAX	2.1265

Note : Replace the camshaft if the outside diameter of any journal is below specification.

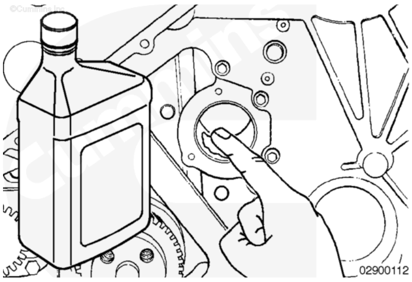
Install

Install the tappets. Refer to Procedure 004-015 in Section 4. (/qs3/pubsys2/xml/en/procedures/100/100-004-015.html)

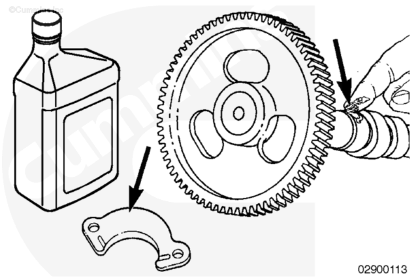


Apply a coat of clean engine oil to the front camshaft bore.

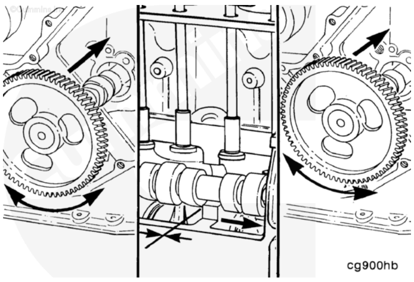
Note : The crankshaft must be positioned so the number 1 cylinder is at approximately top dead center, so the camshaft does not hit the crankshaft counterweight during installation.



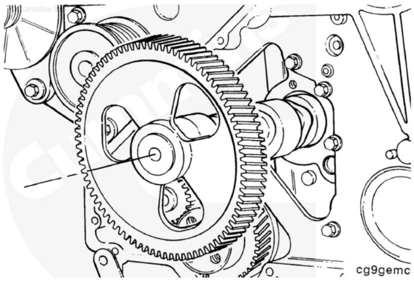
Lubricate the camshaft lobes, journals, and thrust washer with clean engine oil.



Install the camshaft. While pressing in slightly, rotate the camshaft and carefully work the camshaft through the camshaft bushings. As each camshaft journal passes through a bushing, the camshaft will drop slightly and the camshaft lobes will catch on the bushings. Rotating the camshaft will free the lobe from the bushing and allow the camshaft installation to continue.



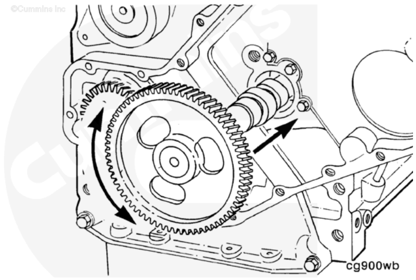
Position the camshaft and gear assembly into the cylinder block up to the last journal.



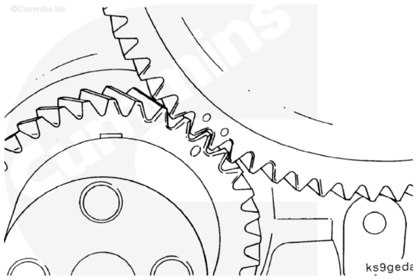
⚠ CAUTION ⚠

Do not try to force the camshaft into the camshaft bore, damage to the camshaft bushing can result.

Before the camshaft gear engages the crankshaft gear, check the camshaft for ease of rotation. When installed properly, the camshaft will rotate freely.



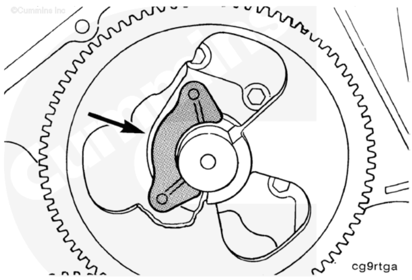
Align the timing marks on the camshaft with the timing marks on the crankshaft.



⚠ WARNING ⚠

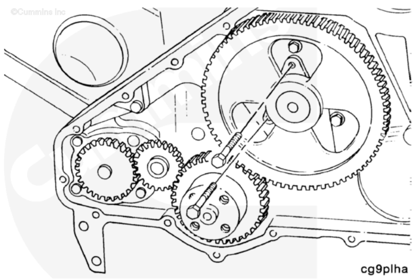
To reduce the possibility of personal injury, make sure the camshaft assembly does not drop on your fingers when installing the thrust plate.

Install the camshaft thrust plate.



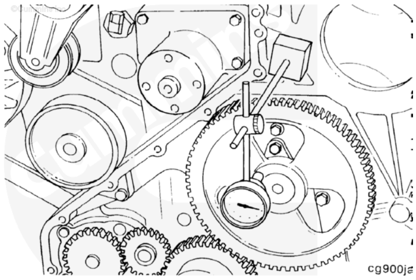
Install and tighten the camshaft thrust plate capscrews.

Torque Value: 24 n•m [212 in-lb]



Measure camshaft end clearance with a dial indicator.

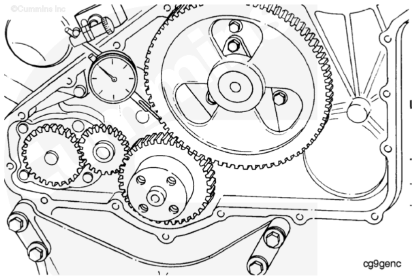
Note : End clearance is controlled by the thickness of the thrust plate and the groove in the camshaft.



Camshaft End Clearance		
mm		in
0.12	MIN	0.005
0.46	MAX	0.018

Note : If the camshaft end clearance is greater than the above limit, the thrust plate must be replaced.

Note : Excessive gear backlash can be caused by worn camshaft bushings.

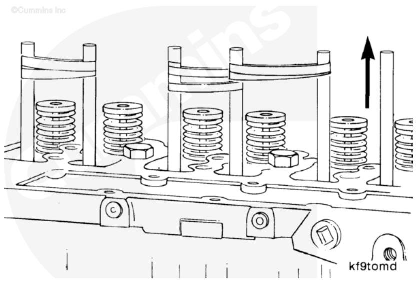


Place a dial indicator on the camshaft gear. Check the gear backlash between the camshaft gear and the crankshaft gear.

Camshaft Gear Backlash Limits (B)		
mm		in
0.08	MIN	0.003
0.33	MAX	0.013

Note : The gears must be replaced if the backlash is greater than the above limit.

Remove the wooden dowel rods from the valve tappets.



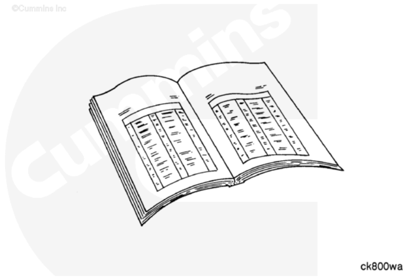
Finishing Steps

Note : Use only Cummins® approved lubricating oil for natural gas engines to prevent premature engine damage.

- Install the front gear cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/100/100-

001-031.html)

- Install the vibration damper. Refer to Procedure 001-052 in Section 1. (/qs3/pubsys2/xml/en/procedures/100/100-001-052.html)
- Install the drive belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/100/100-008-002.html)
- Install the push rods. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/100/100-004-014.html)
- Install the rocker levers. Refer to Procedure 003-008 in Section 3. (/qs3/pubsys2/xml/en/procedures/100/100-003-008.html)
- Adjust the valve lash. Refer to Procedure 003-004 in Section 3. (/qs3/pubsys2/xml/en/procedures/41/41-003-004.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/100/100-003-011.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/100/100-008-018.html)
- Fill the engine oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-037.html)
- Operate the engine until the coolant temperature reaches 82° C [180° F]. Check for leaks and proper operation.



Last Modified: 02-Oct-2017